META RADAR

From Scattered Signals to Strategic Intelligence

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Problem Statement #3: From Inbox Noise to Strategic Signal | **Pilot Domain:** Haemophilia A & B (Rare Disease)

1. The Problem

The haemophilia treatment landscape is rapidly evolving (non-factor therapies *emicizumab*, *concizumab*, *mim8*; AAV gene therapies *Hemgenix*, *Roctavian*). Critical external signals are fragmented across 8 silos: pharma news, clinical trial registries, scientific publications, medical congresses (ASH, ISTH, WFH), regulatory releases, safety signals, patient/access narratives, and competitor pipeline activity. Cross-functional teams manually find, connect, and interpret these updates. The challenge is turning fragmented external information into actionable intelligence for the right team.

Scattered External Sources ⟶ Understand ⟶ Prioritize ⟶ Route ⟶ Act

2. External Signal Sources

MetaRadar monitors a multi-source data landscape categorized across six domains. Primary sources are supported through API connectors in the prototype, supplemented by simulated feeds for extended categories:

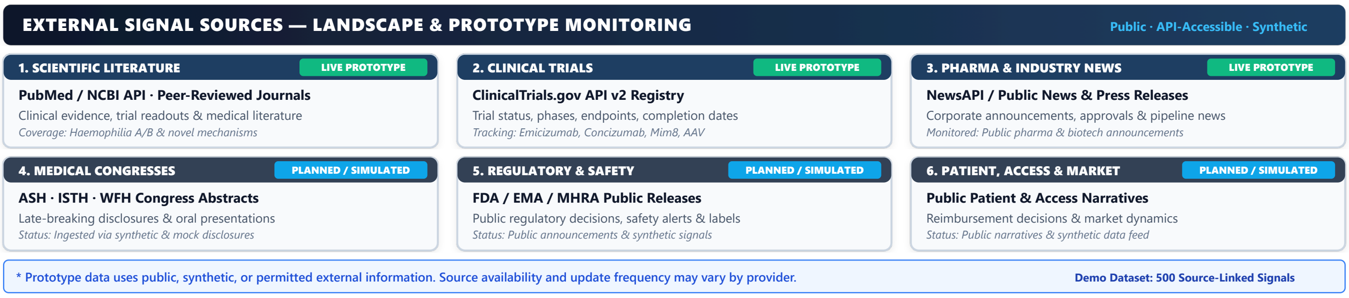

Figure 1: External Signal Landscape & Scope — Prototype live connectors explicitly distinguished from planned categories.

3. Our Solution

MetaRadar is an AI-powered intelligence radar that converts fragmented external haemophilia updates into source-linked, prioritized, explainable, and actionable intelligence. Rather than operating as a generic news dashboard, MetaRadar connects evidence around distinct developments and delivers function-specific insights to six primary Novo Nordisk stakeholders: **Medical Affairs, Regulatory, Safety** / **Pharmacovigilance, Market Access, Medical Communications, and Leadership**. Importantly, not every signal is sent to every function; MetaRadar routes each development to the teams most relevant to it.

4. The Four Business Questions

MetaRadar structures intelligence around four decision-first questions:

Business Question	Executive Intelligence Framing
1. WHAT CHANGED?	Factual summary grounded in relevant external sources.
2. WHY DOES IT MATTER?	Why it matters, supported by evidence and checks for conflicting information.
3. WHO SHOULD REVIEW IT?	Calibrated routing to the most relevant Novo Nordisk function.
4. WHAT ACTION MAY BE REQUIRED?	Suggested next steps for the relevant team, subject to human review.

5. High-Level Processing Flow

MetaRadar transforms incoming external information through a human-readable six-stage workflow:

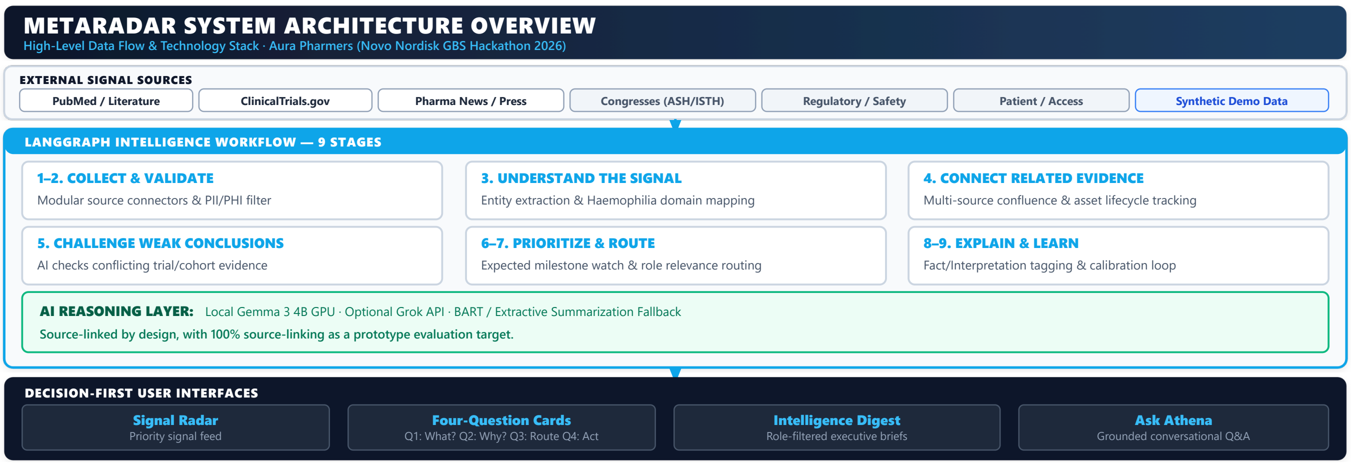

Figure 2: MetaRadar System Architecture — High-level data flow from public and synthetic sources through intelligence workflow and UI.

6. What Makes MetaRadar Different?

MetaRadar is distinguished from conventional monitoring tools by **five core intelligence mechanisms**:

1. **CONFLUENCE:** Connects independent updates pointing to the same development within a rolling 48-hour window, so teams see one stronger intelligence story instead of several disconnected alerts.
2. **LIFECYCLE:** Shows how a development evolves over time across a clear progression: *Announcement* → *Clinical Trial* → *Congress Readout* → *Publication* → *Regulatory Decision* → *Access Event* → *Post-Market Evidence*.
3. **RED-TEAM:** Challenges weak or unsupported AI interpretations by looking for conflicting evidence, population differences, endpoint mismatches, and other limitations that could change the interpretation.
4. **WATCH-FOR-NEXT:** Tracks developments that may require follow-up and watches for important future evidence or expected milestones, such as congress abstracts or regulatory developments.
5. **STAKEHOLDER CALIBRATION:** Compares AI baselines with expert feedback so relevance, urgency, routing, and suggested actions become more accurate and useful over time.

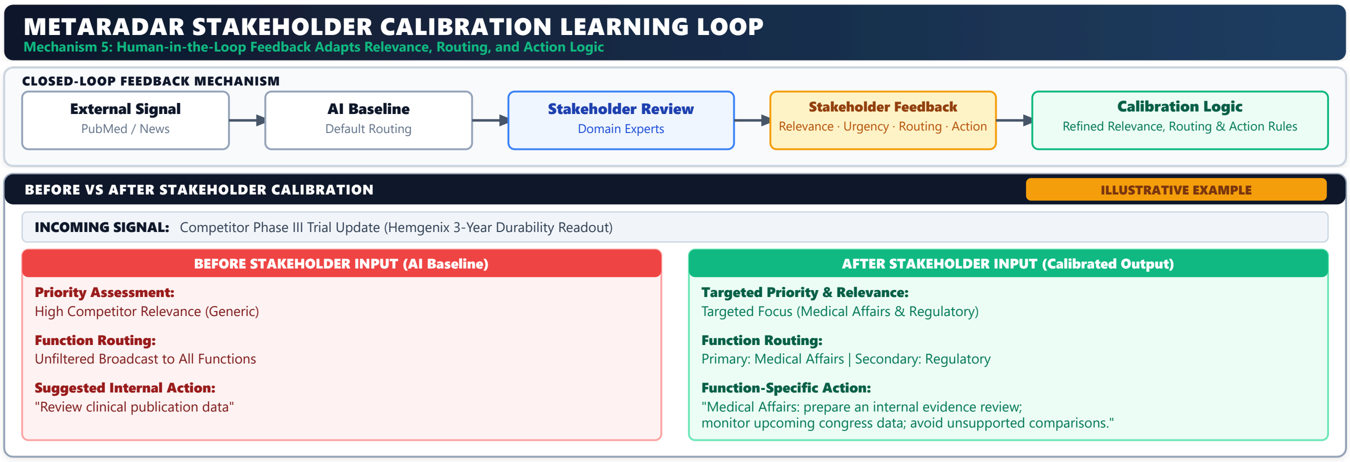

Figure 3: Stakeholder Calibration Learning Loop — Before vs. after stakeholder feedback for a competitor Phase III trial update (*Illustrative Example*).

7. Technology at a Glance

Category	Primary Component	Secondary / Hosted	Fallback / Utility
AI Reasoning	Gemma 3 4B (Local GPU)	xAI Grok API (Optional)	BART / Extractive Summarization Fallback
Intelligence	Entity extraction & evidence linking	Confluence & lifecycle analysis	Red-Team & stakeholder calibration
Platform & Infra	LangGraph / FastAPI	Next.js 15 UI	PostgreSQL / Redis 7
Data Sources	PubMed Literature	ClinicalTrials.gov	NewsAPI / Regulatory

8. Safe by Design

- Public, mock, or synthetic data only (zero confidential Novo Nordisk data and zero patient-identifiable PII/PHI).
- Every important insight remains linked to its supporting external source, evidence, and date.
- Each signal retains its source, publication/update date, and supporting evidence so users can trace an insight back to the underlying information.
- **FACT / INTERPRETATION / SPECULATION** are explicitly separated so assumptions are not presented as facts.
- Confidence reflects evidence strength, not medical certainty.
- AI suggests; human experts review and decide. High-impact safety, regulatory, and medical outputs require human review.

9. Designed to Scale

Haemophilia is the pilot, not the limit of the platform. The core MetaRadar engine is separated from disease-specific knowledge, so another therapy area can reuse the same intelligence workflow while changing the relevant disease knowledge, terminology, sources, and routing rules.

10. Prototype Evaluation & Expected Outcome

MetaRadar is evaluated against verifiable hackathon targets: **100% source-linked summaries**, **$\geq$85% classification accuracy**, **$\leq$5-minute discovery time** for top signals, **zero confidential/patient data**, and **demonstrable calibration improvement**.

Expected Outcome: MetaRadar transforms scattered external noise into clear strategic signal, empowering Novo Nordisk cross-functional teams to identify key developments faster, understand their significance, route them to the right function, and take informed internal action with confidence.